

1 Fig. 1.1 is a diagram showing a stage in protein synthesis.

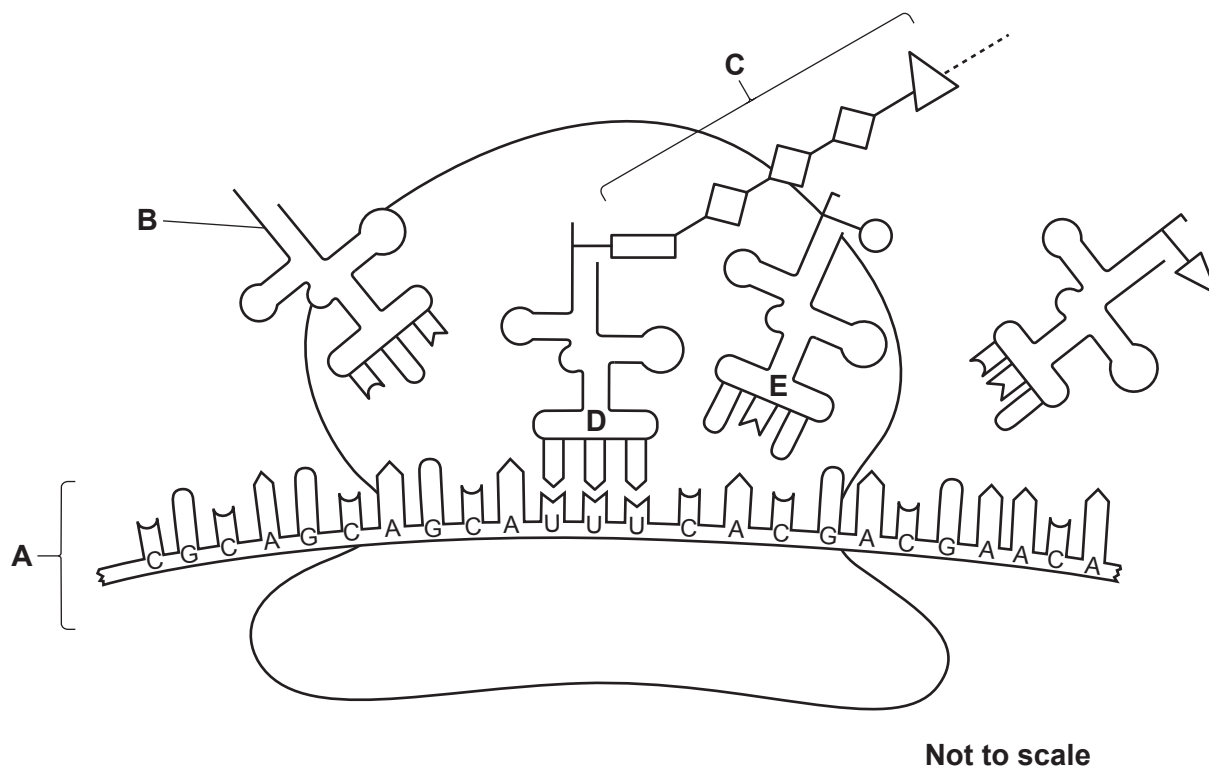


Fig. 1.1

(a) (i) Name the stage of protein synthesis that is shown in Fig. 1.1.

..... [1]

(ii) Identify **A**, **B** and **C** in Fig. 1.1.

A

B

C

[3]

(iii) State the base sequences at **D** and **E**.

D

E

[1]

- (b) Mutagenesis is a process that leads to a change in the amino acid sequences of proteins. Scientists carry out mutagenesis to investigate the importance of particular amino acids in protein structure and function.

Outline how changing one amino acid in the β -globin polypeptide of haemoglobin may change the structure and function of a molecule of haemoglobin.

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..... [4]

[Total: 9]